2016 EXAMINATIONS

Lancaster University

Part II

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.361 Artificial Intelligence [2.5 Hours]

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1a Describe the three main types of machine learning methods and give an example for each one of them. **[6 marks]**

1b Describe the 5 levels of autonomy including the sub-levels for levels 4 and 5. **[5 marks]**

1c For the following two data samples:

$x_1=\{4; 2; 1; 9\}$

$x_2=\{160; 420; 1010; 10\}$

i. calculate the range and mean; **[4 marks]**

ii. perform normalization; **[2 marks]**

1d Describe the procedure of the *k*-means clustering algorithm. Illustrate this with a simple example for two iterations using $k=2$ and the following *1D* (one dimensional) data set which represents the temperature $T=\{10; 12; 8; 4; 16; 17\}$.

[8 marks]

Total 25 marks

Question 2

2.a Sketch and describe a radial-basis function (RBF) type artificial neural network (ANN).

[5 marks]

2.b The following table shows the expected profit of three investment decisions made by an investment banker under different conditions (future oil price changes). Using Maximax, Maximin, Lagrangian (equal probabilities), and Minimum Regret Decision Making methods determine which investment is best to be made and what it will bring to the investment banker.

Future oil price changes	**Expected Profit**		
	Transport & Travel	Manufacturing	Agriculture
Decrease	*£90M*	*£15M*	*£50M*
Stable	*£25M*	*£30M*	*£50M*
Increase	*£8M*	*£50M*	*£40M*

[8 marks]

2.c Compare a fuzzy set with a normal set and give an example for each one.

[3 marks]

2.d Which type of Fuzzy Rule-based System is associated with each of the following IF THEN rules and why:

i. *$Rule_1$: IF (House_Area is **Large**) AND (Location is **Attractive**) AND...*
*THEN (Cost is **High**)*

[3 marks]

ii. *$Rule_1$:* IF (*House_Area is **Large**) AND (Location is **Attractive**)*
THEN (Cost=£200K(0.1+0.3 House_Area +0.6* Location)*

[3 marks]

iii. Compare the two types of Fuzzy Rule-based models in terms of their antecedent (IF) part and their consequent (THEN) part.

[3 marks]

Total 25 marks

Question 3

3a This question is about searching algorithms.

For questions (i-iii), define the following terms in the state space representation context

i. State space (1 mark).

ii. Action (1 mark).

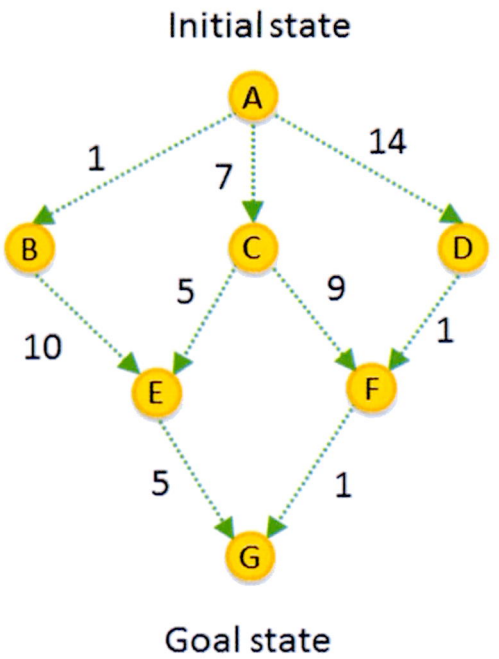
iii. Completeness of a search algorithm (2 marks).

iv. Define Breadth-First Search (BFS) and Depth-First search (DFS) (2 marks). Compare these two algorithms in terms of the completeness and optimality (2 marks).

v. In a uniform-cost search, each action is associated with a cost. The graph given below shows a search space that consists of 7 states {A,B,C,D,E,F,G} and the actions joining pairs of states. The weight of an edge represents the cost of the corresponding action. Given the initial state (node) A, provide the sequence of nodes evaluated in a uniform-cost search before it identifies the best path from A to the goal state G: (4 marks)

[Note: The answer to this question is not to enumerate all nodes in the best path from A to G, but the sequence of nodes that are evaluated by the uniform cost search. Some nodes in this sequence may not be contained in the best path.]

[A, ?, ?, ?, ... G]



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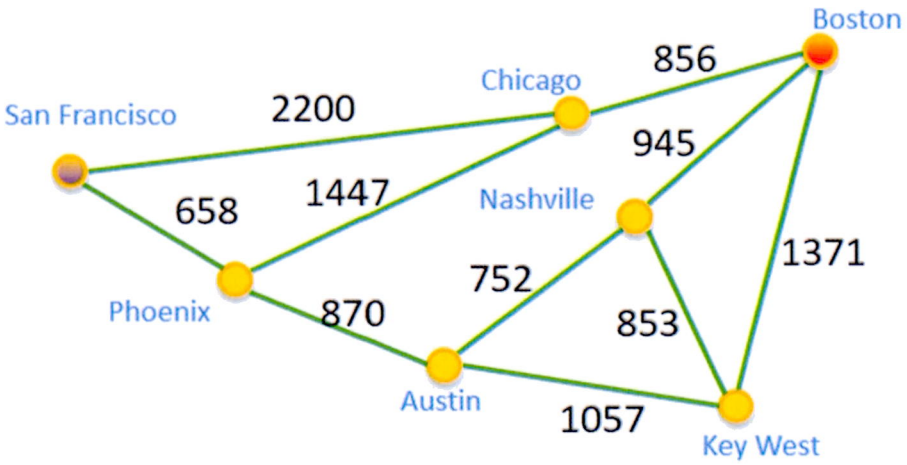
Question 3 continued…

vi. Define an admissible heuristic (function) (2 marks).

vii. A* search combines greedy search and uniform cost search based on heuristic functions. Given a search space that consists of 7 cities and the cost of the action associated with each edge (the distance; shown in the figure given below), provide the sequence of nodes evaluated by A* search before it finds the best path from the initial state Boston to the goal state San Francisco. The heuristic function *h* is given below in the table: (5 marks)

[Note: Similarly to v, the answer to this question is not the node sequence in the best path from Boston to San Francisco, but the sequence of nodes that are evaluated by the uniform cost search]

[Boston, ?, ?, ?, …. San Francisco]



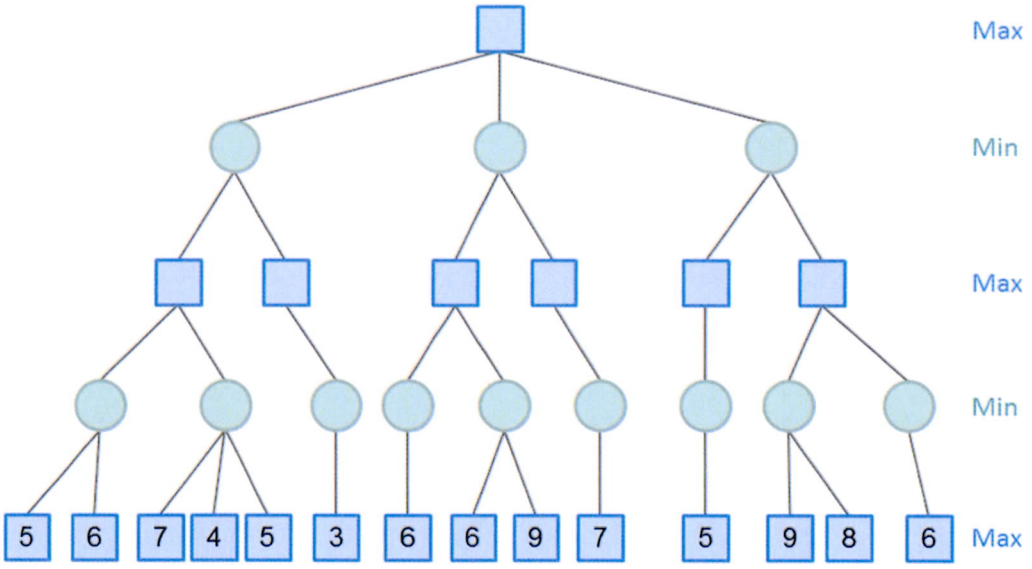
Node	*h* (cost. San Francisco)
Boston	2500
Chicago	2200
Nashville	2100
Key West	2400
Austin	1450
Phoenix	658

[19 marks]

Question 3 continues on next page…

Question 3 continued…

3b Apply alpha-beta pruning to the search tree shown below. Indicate all parts of the tree that are not evaluated (4 marks). Provide the evaluated value of the root node (2 marks).



[6 marks]

Total 25 marks

Question 4

4a This question is about the entropy of a random variable. Suppose a random variable x takes values in S = {0,1,2}. Provide the probability table (the values of x and the corresponding probabilities) which leads to the highest and the lowest entropy of x.

Highest entropy case: (3 marks)

$x = 0$	$x = 1$	$x = 2$
?	?	?

Lowest entropy case: (3 marks)

$x = 0$	$x = 1$	$x = 2$
?	?	?

[6 marks]

4b This question is about applying the Bayes' rule.

Suppose on average,
1 in 100 patients reports a stiff neck,
1 in 10000 patients has meningitis,
and meningitis causes a stiff neck 40% of the time.

In this case, if a patient has a stiff neck, what is the probability that the patient has meningitis? Explain your answer.

[5 marks]

4c This question is about genetic operators in Genetic Algorithms (GAs).

Suppose that we are given a traveling salesman problem where we have a list L of 8 cities {A,B,C,D,E,F,G,H} and the cost of travel between each pair of cities. The goal is to find the least costly route that visits each city exactly once.

This problem can be approached using the GAs with permutation encoding:
A chromosome (a representation of a hypothesised solution in GAs) represents an ordering of places, e.g.,

Chromosome1 = [A, B, C, D, E, F, G, H]
Chromosome2 = [C, B, E, G, H, F, D, A]

Question 4 continues on next page...

Question 4 continued...

Show the result (two output chromosomes) of applying one point crossover to Chromosome1 and Chromosome2. The crossover point is between the third and the fourth genes in each chromosome:

Output chromosome1 = [?, ?, ?, ?, ?, ?, ?, ?]
Output chromosome2 = [?, ?, ?, ?, ?, ?, ?, ?]

[5 marks]

4d This question is about the decision tree. Take the dataset below that consists of twelve examples (X1 through X12) with two binary attributes (A and B). Each example is provided with corresponding binary class labels (-1 or 1).

Dataset

	A	B	Class label
X1	1	1	1
X2	0	1	1
X3	0	0	-1
X4	1	1	1
X5	0	1	-1
X6	1	0	-1
X7	1	0	-1
X8	0	0	1
X9	0	0	1
X10	1	1	-1
X11	1	1	-1
X12	0	1	1

i. Calculate the entropy H(Class label) of variable `Class label'. You need to show your calculations up to the log2() expressions (i.e. you do not have to evaluate the log2() expressions). (3 marks)

H(Class label) = ?

Question 4 continues on next page...

Question 4 continued…

ii. Now assume that you have chosen attribute B for the root node of the decision tree.

- Calculate the entropy H(Class label) of variable `Class label' when B = 1 (1 mark):

H(Class label) when (B=1) = ?

- Similarly, calculate the entropy H(Class label) of variable `Class label' when B = 0 (1 mark):

H(Class label) when (B=0) = ?

- Based on these two quantities, calculate the average entropy H(Class label) given attribute B. The average entropy H(Class label) given attribute B will be denoted as `Avg[B](Class label)' (2 marks):

Avg[B](Class label) = ?

- Using Avg[B](Class label), calculate the information gain IG(B) of attribute B. If you did not calculate Avg[B](Class label), you can simply refer the quantity as `Avg[B](Class label)' (2 marks):

IG(B) = ?

[9 marks]

Total 25 marks

---End of Paper---